

AYE
TECH HUB

• MEP ENGINEERING • LESSON 01

Introduction to MEP Engineering

Lesson 01 of 30 | MEP Engineering Program

Awet G. Nway

Founder, AYE Tech Hub | MEP & Building Services Engineer

- What MEP is and why it accounts for 40-60% of construction cost
- The three disciplines: Mechanical, Electrical and Plumbing
- MEP in the building construction process — 6 lifecycle phases
- Key roles: consultant, coordinator, site supervisor, commissioning
- International codes (ASHRAE, IEC, NFPA, IPC) and career pathways

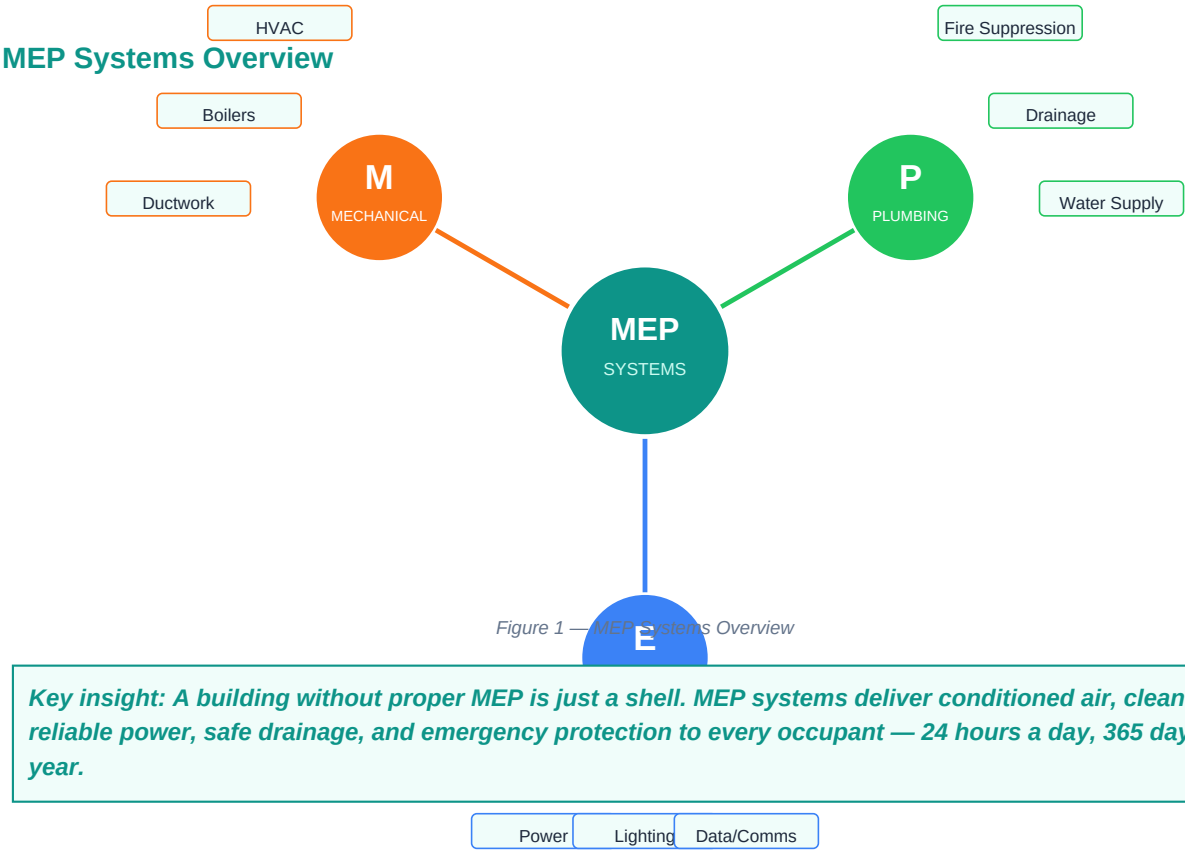
What is MEP Engineering?

MEP stands for **Mechanical, Electrical, and Plumbing** — the three core building services disciplines that make a structure functional, safe, and comfortable for occupants. While architects design the form and structural engineers design the frame, MEP engineers design the systems that allow people to live and work inside a building.

MEP systems account for **40–60% of total building construction cost** on commercial and institutional projects. They are designed in parallel, coordinated together, and installed before finishes. Poor MEP coordination is one of the leading causes of construction rework and cost overrun globally.

The Three Disciplines at a Glance

Discipline	Covers	Key Systems
Mechanical (M)	Thermal comfort, air quality, fluid movement	HVAC, boilers, chillers, AHUs, ductwork, piped services
Electrical (E)	Power, light, communications, safety systems	LV/MV distribution, lighting, UPS, fire alarm, data/comms
Plumbing (P)	Water supply, drainage, fire suppression	Cold/hot water, drainage, sewage, sprinklers, medical gas



MEP in the Construction Process

MEP engineers are involved from early feasibility right through to building handover and operational support. The earlier MEP design starts, the fewer costly changes occur on site. On large projects, MEP subcontractors may represent three separate specialist companies coordinated by an MEP lead consultant.

MEP Project Lifecycle



Figure 2 — MEP Project Lifecycle Phases

Phase	MEP Activity	Key Deliverable
Feasibility	Building services strategy, energy concept, load estimates	Services concept report, plant room sizing
Schematic Design	System selection, spatial allocation, riser/zone planning	Schematic drawings, outline specification
Detailed Design	Equipment sizing, pipe/duct routing, full load calculations	Construction drawings, BOQ, specifications
Procurement	Tender packages, equipment approval, shop drawing review	Approved submittals, manufacturer data sheets
Installation	Site supervision, coordination with structure & architecture	Progress reports, as-built markups
Testing & Commissioning	System testing, TAB (test, adjust, balance), performance verification	Test reports, O&M; manuals, training records

Key MEP Roles and Responsibilities

Role	Primary Responsibility
MEP Lead Consultant	Overall design coordination, client interface, code compliance, sign-off
Mechanical Engineer	HVAC, boilers, chillers, ductwork, piping, thermal load calculations
Electrical Engineer	Power distribution, lighting, fire alarm, earthing, data/comms design
Plumbing Engineer	Water supply, drainage, fire protection, sanitary, gas systems
MEP BIM Coordinator	Clash detection, 3D coordination, IFC model management, issue tracking
MEP Site Supervisor	Installation oversight, subcontractor management, quality inspection
Commissioning Engineer	System testing, performance balancing, handover documentation

Career tip: Junior MEP engineers typically specialise in one discipline (M, E, or P) first and gradually build cross-discipline knowledge. Senior roles like MEP Manager or Building Services Director require fluency in all three.

MEP Scope by Discipline

Each discipline has a clearly defined scope of work. Understanding where one discipline ends and another begins is critical for tendering, coordination, and contractual responsibility. The diagram below shows the standard scope breakdown used on most commercial and institutional building projects.

MECHANICAL	ELECTRICAL	PLUMBING
■ HVAC Systems	■ Power Distribution	■ Cold & Hot Water
■ Boilers & Chillers	■ Lighting & Controls	■ Drainage & Waste
■ Ductwork & AHUs	■ Emergency/UPS	■ Sanitary Systems
■ Ventilation (FAU)	■ Earthing & Bonding	■ Fire Sprinklers
■ Piped Services	■ Fire Alarm System	■ Medical Gas
■ Thermal Insulation	■ Data & Comms	■ Rainwater Harvesting

Figure 3 — MEP Scope by Discipline

Key International MEP Codes and Standards

Standard	Discipline	Scope
ASHRAE 90.1	M + E	Energy efficiency standard for buildings — HVAC & lighting
ASHRAE 62.1	Mechanical	Ventilation rates for acceptable indoor air quality (IAQ)
IEC 60364	Electrical	Low-voltage electrical installations in buildings
NFPA 70 (NEC)	Electrical	US National Electrical Code — wiring methods and safety
IPC	Plumbing	International Plumbing Code — water supply and drainage
NFPA 13	P + Fire	Installation standard for automatic sprinkler systems
EN 12831 / BS EN	Mechanical	Design heating load calculation method for buildings
CIBSE Guides	M + E + P	UK/international best-practice guides for building services

MEP Career Pathways

Level	Typical Title	Experience & Focus
Entry	Graduate / Junior MEP Engineer	0–2 yrs — learning software (Revit MEP, AutoCAD MEP), assisting design
Mid	MEP / Design Engineer	2–6 yrs — leading system design, producing coordinated drawings, site visits
Senior	Senior Engineer / Project Engineer	6–10 yrs — project ownership, client coordination, specification writing
Lead	MEP Manager / Principal Engineer	10+ yrs — multi-project leadership, fee proposals, team management
Director	Head of MEP / Building Services Director	15+ yrs — business strategy, P&L; responsibility, large programme delivery

What comes next — MEP-002: How MEP systems are integrated into the building construction workflow — coordination drawings, MEP zones, service corridors, riser shafts, and clash detection using BIM.